SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Miranalini Joys (192521054) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks /**26**, /**27**, and /**28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I J.P.MIRUNLINI JOYS (192521054) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
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Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:
School of Computing → /25
School of Electronics → /26
School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

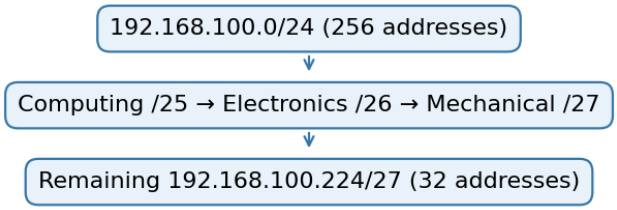
ANSWER 1 – SMART CAMPUS SUBNET DESIGN

Scenario and key issues: The 192.168.100.0/24 block has 256 addresses. Three different fixed prefix lengths require VLSM allocation in descending block size; subnets must not overlap and network/broadcast addresses are reserved.

Method: Total addresses = 2^(32 − prefix); usable IPv4 hosts = 2^(32 − prefix) − 2. /25: 128 − 2 = 126; /26: 64 − 2 = 62; /27: 32 − 2 = 30.

Department	Network / mask	Usable host range	Broadcast	Hosts
Computing	192.168.100.0/25	192.168.100.1 – 192.168.100.126	192.168.100.127	126
Electronics	192.168.100.128/26	192.168.100.129 – 192.168.100.190	192.168.100.191	62
Mechanical	192.168.100.192/27	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Allocated addresses = 128 + 64 + 32 = 224. Unallocated = 256 − 224 = 32 addresses, spanning 192.168.100.224 – 192.168.100.255 (available /27). Total usable hosts across the three subnets = 126 + 62 + 30 = 218.



Decision and conclusion: Allocate largest blocks first to ensure alignment and avoid overlap. Separate departmental VLANs and access rules improve management and security; reserve the unused /27 for expansion.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:
Software Development → **/24**
Quality Assurance → **/25**
Human Resources → **/26**
Executive Management → **/27**
As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWER 2 – SOFTWARE COMPANY VLSM

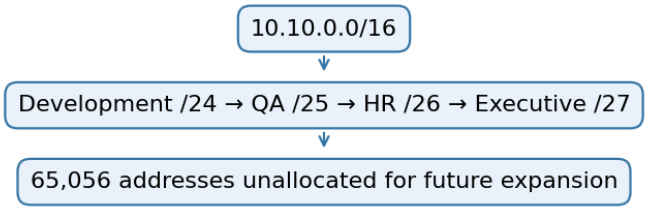
Scenario and key issues: 10.10.0.0/16 contains 65,536 addresses. Departmental subnet sizes differ; allocate in descending size without overlaps and preserve unused space.

Method: VLSM; usable hosts = 2^(32 − prefix) − 2. /24 = 254; /25 = 126; /26 = 62; /27 = 30.

Department	Network	Valid host range	Broadcast	Hosts
Software Development	10.10.0.0/24	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129 – 10.10.1.190	10.10.1.191	62

Executive Management	10.10.1.192/27	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Allocated = 256 + 128 + 64 + 32 = 480 addresses. Remaining = 65,536 − 480 = 65,056 unallocated addresses. Unused range: 10.10.1.224 – 10.10.255.255. Total usable hosts in allocated subnets = 254 + 126 + 62 + 30 = 472.



Decision and conclusion: VLSM matches each department's allocated size, preserves future capacity and allows departmental VLAN separation. HR and executive networks should have controlled interdepartmental access.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER 3 – SMART HOSPITAL IPv6

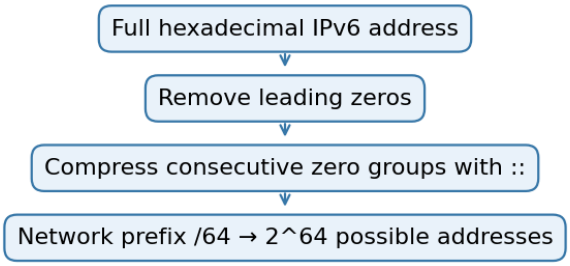
Scenario and key issues: IPv6 uses 128-bit addresses. The /64 prefix identifies 64 network bits and leaves 64 interface-identifier bits. Leading zero suppression and :: compression simplify notation.

Given: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 1: Remove leading zeros → 2001:db8:abcd:0:0:0:0:0/64. Step 2: Replace the five consecutive all-zero groups with :: → 2001:db8:abcd::/64.

Expanded full address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Network prefix: 2001:db8:abcd::/64. First 64 bits: 2001:0db8:abcd:0000; remaining 64 bits: 0000:0000:0000:0000.

Total possible addresses = 2^(128 − 64) = 2^64 = 18,446,744,073,709,551,616 (approximately 18.45 quintillion). The question's two host-address calculations both refer to this /64 address capacity; actual assignable unicast addresses depend on reserved/special-purpose rules. IPv6 has no broadcast address.



Decision and conclusion: /64 is a conventional IPv6 LAN prefix suitable for SLAAC. Separate medical equipment, administrative and guest traffic for security. Note: 2001:db8::/32 is a documentation-only prefix, not a production allocation.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
A packet arrives at the router with the destination IP address:
192.168.2.45
As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

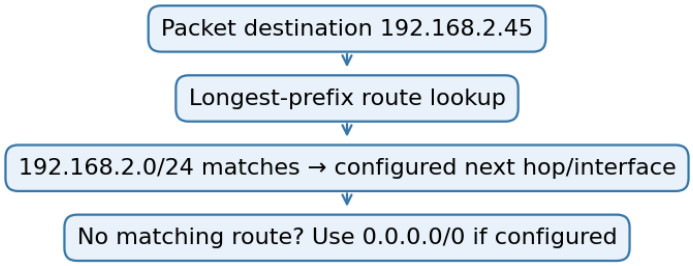
ANSWER 4 – ISP ROUTING DECISION

Scenario and key issues: The router has routes to 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. The destination is 192.168.2.45. Routing uses longest-prefix matching.

Calculation: /24 corresponds to 255.255.255.0. 192.168.2.45 belongs to 192.168.2.0/24 (host range 192.168.2.1 – 192.168.2.254; broadcast 192.168.2.255).

ROUTING ENTRY	MATCH FOR 192.168.2.45?
192.168.1.0/24	No
192.168.2.0/24	Yes – selected
192.168.3.0/24	No

Next hop/interface: Forward via the next-hop IP or outgoing interface configured for 192.168.2.0/24. The provided routing entries contain only destinations, so an exact interface name or next-hop IP cannot be determined. If no specific route matches, use default route 0.0.0.0/0 if configured; otherwise discard the packet.



Decision and conclusion: The matching route is 192.168.2.0/24. Longest-prefix matching sends the packet along the most specific available path; the default route is a fallback only.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:
Administration LAN → **192.168.10.0/26**
Finance LAN → **192.168.10.64/26**
The router interfaces are configured with:
Administration Gateway → **192.168.10.1**
Finance Gateway → **192.168.10.65**
As the network administrator, perform the following tasks:

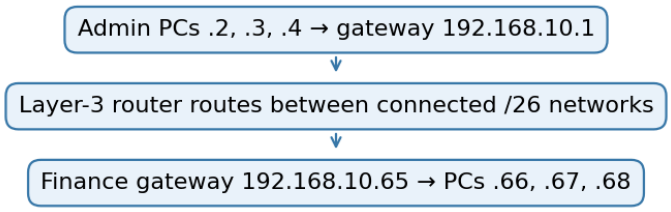
1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWER 5 – TWO LANs WITH A LAYER-3 ROUTER

Scenario and key issues: Two distinct /26 networks connect through a router. Both use subnet mask 255.255.255.192; each /26 has 2^6 = 64 total addresses and 64 − 2 = 62 usable host addresses. Gateway addresses must remain unique and within their LANs.

LAN	Network	Valid host range	Broadcast	Gateway
Administration	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127	192.168.10.65

Computer	IP address	Subnet mask	Default gateway
Admin PC1	192.168.10.2	255.255.255.192	192.168.10.1
Admin PC2	192.168.10.3	255.255.255.192	192.168.10.1
Admin PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance PC1	192.168.10.66	255.255.255.192	192.168.10.65
Finance PC2	192.168.10.67	255.255.255.192	192.168.10.65
Finance PC3	192.168.10.68	255.255.255.192	192.168.10.65



Decision and conclusion: Use the given router gateways and assign PCs from valid, unused host ranges. An Admin PC sending to Finance forwards to 192.168.10.1; the router forwards via the Finance interface. Successful communication requires working links and permitted security policies.

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of	25%	Applies appropriate	Applies concepts	Limited or partially	Unable to apply

Concepts		concepts effectively to address the scenario	with minor errors	correct application	relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand